

Level-Two Stable Rationality for Quartic del Pezzo Surfaces

Tavis Rudd*

August 2026

Abstract

Let k be a field of characteristic zero and let S/k be a smooth quartic del Pezzo surface. We prove that if $S(k) \neq \emptyset$ and $\text{Pic}(S_{\bar{k}})$ is a stably permutation Galois lattice, then $S \times \mathbf{A}^2$ is k -rational. In particular every stably rational quartic del Pezzo surface has stable-rationality level at most two. The proof uses the Cox compactification of a universal torsor. A saturated rank-three subtorus has four surviving Cox-weight blocks whose weights form a unimodular tetrahedron. A codimension-three tangent section of the OADP compactification meets each general orbit once; the residual torus has rank two and is rational. This is an instance of a general unimodular-window quotient theorem for torus actions on varieties with one apparent double point. Applied to the two cubic families of Tschinkel and Zhang, the result gives $X \times \mathbf{P}^2$ rational in every dimension in which their construction applies. For their cubic threefolds, the known irrationality of $X \times \mathbf{P}^1$ makes the stabilization threshold exactly two and gives smooth projective nonrational fourfolds Y for which $Y \times \mathbf{A}^1$ is rational.

Keywords. quartic del Pezzo surface; stable rationality; universal torsor; OADP variety; algebraic torus; cancellation.

1 Results and mechanism

For a stably rational variety V/k , write

$$s(V) = \min\{m \geq 0 : V \times \mathbf{P}^m \text{ is } k\text{-rational}\}.$$

The value depends on the ground field. Our main result gives a uniform upper bound for quartic del Pezzo surfaces.

Theorem 1.1 (Level two for quartic del Pezzo surfaces). *Let k be a field of characteristic zero and let S/k be a smooth quartic del Pezzo surface. Suppose that $S(k) \neq \emptyset$ and that $\text{Pic}(S_{\bar{k}})$ is a stably permutation Galois lattice. Then*

$$S \times \mathbf{A}^2$$

is k -rational. Consequently every stably k -rational smooth quartic del Pezzo surface satisfies the same conclusion.

The mechanism has two parts. Tschinkel and Zhang prove that a universal torsor with a rational point is rational by compactifying it to an OADP variety, that is, a projective variety through whose general ambient point passes one secant line [8, Theorem 3.4]. We use the same compactification equivariantly. A rank-three subtorus leaves four Cox-weight blocks outside a fixed boundary space. Their weights are the vertices of a unimodular tetrahedron. Three tangent equations therefore select one point of each general orbit. The quotient is rational, and the remaining torus has dimension two.

*tavisrudd@damnsimple.com

Corollary 1.2 (The Tschinkel–Zhang cubic families). *Every cubic hypersurface $X/\mathbf{Q}$ in the two families of Tschinkel–Zhang Propositions 5.1 and 5.2 satisfies*

$$X \times \mathbf{P}^2 \text{ is rational over } \mathbf{Q}.$$

For the two cubic threefolds this determines the exact level. The equations are

$$X_1 = \{(x_4 - 2x_3)x_1^2 + 3(x_4 + 2x_3)x_2^2 + 3x_3^2x_4 - x_4^3 + x_5^3 = 0\}, \quad (1)$$

$$X_3 = \{x_4(x_1^2 + 2x_1x_2) + x_3(x_1^2 + x_1x_2 + x_2^2) + x_3^3 - x_3^2x_4 + x_4^3 + 2x_5^3 = 0\}. \quad (2)$$

Corollary 1.3 (Exact threshold and cancellation). *For either smooth cubic threefold $X = X_1, X_3 \subset \mathbf{P}^4$,*

$$s(X) = 2.$$

Moreover $Y = X \times \mathbf{P}^1$ is a smooth projective nonrational fourfold over $\mathbf{Q}$, while $Y \times \mathbf{A}^1$ is $\mathbf{Q}$ -rational.

Tschinkel and Zhang obtain stable rationality from rational universal torsors and stable-permutation Picard lattices. Theorem 1.1 uses their universal-torsor theorem and their four-type classification; it sharpens the resulting stabilization bound. Thus their geometric input is load-bearing rather than a consequence of the argument here.

2 A unimodular-window quotient theorem

Let $Z \subset \mathbf{P}(V)$ be a geometrically integral n -fold over an infinite field k . We say that Z has the tangent-projection property if projection from T_pZ at a general $p \in Z(k)$ is birational to $\mathbf{P}^n$. An OADP n -fold in $\mathbf{P}^{2n+1}$ with dense k -points has this property [8, Theorem 2.4].

Theorem 2.1 (Unimodular-window quotient). *Let a rank- r k -torus T act linearly and generically freely on a variety $Z \subset \mathbf{P}(V)$ with the tangent-projection property. Over k^s , suppose that V has nonzero weight blocks*

$$V_{w_0}, \dots, V_{w_r}$$

such that $w_1 - w_0, \dots, w_r - w_0$ is a $\mathbf{Z}$ -basis of $X^(T_{k^s})$. Suppose also that a k -defined linear space B contains every other weight block. Assume that for a general $p \in Z(k)$ there is a k -defined codimension- r linear space*

$$\Lambda \supset \langle T_pZ, B \rangle$$

whose equations restrict to an r -by- $(r+1)$ coefficient matrix of rank r with every coordinate of its kernel nonzero. Then the Rosenlicht quotient Z/T is k -rational.

Proof. Tangent projection maps $\Lambda \cap Z$ birationally to $\mathbf{P}^{n-r}$. On a general geometric orbit, the equations of Λ involve only the $r+1$ surviving characters. Their coefficient matrix fixes every ratio $\chi^{w_j - w_0}(t)$. Since the weight differences form an integral basis, these ratios are coordinates on T_{k^s} ; hence the slice contains exactly one point of the open orbit. The construction is defined over k , so this geometric point is the generic point of a degree-one rational slice over k . Therefore $\Lambda \cap Z \dashrightarrow Z/T$ is birational.

The orbit correction is explicit. Let $\lambda_1, \dots, \lambda_r$ be equations of Λ , write q_j for the component of q in V_{w_j} , and set

$$A(q)_{ij} = \lambda_i(q_j), \quad \kappa_j(q) = (-1)^j \det A(q)_{\widehat{j}}.$$

Cramer's identity and the nonvanishing of the maximal minors give the unique element $t(q)$ through

$$\chi^{w_j - w_0}(t(q)) = \frac{\kappa_j(q)}{\kappa_0(q)}, \quad 1 \leq j \leq r.$$

No root extraction occurs. Uniqueness makes the formula Galois equivariant, even when the individual weight blocks do not descend. The map $q \mapsto t(q)q$ is the rational retraction to the slice. $\square$

The rank-one case says that two adjacent weights suffice. The higher-rank form explains the role of saturation: a finite-index simplex would produce a finite cover of the orbit rather than a one-point slice.

3 The saturated rank-three window

Let $\mathcal{T} \rightarrow S$ be a universal torsor and T_0 the rank-five projective Neron–Severi torus obtained after quotienting the anticanonical scalar. Over $\bar{k}$, the projective Cox closure

$$Z = \mathbf{P}(\text{Spec Cox}(S)) \subset \mathbf{P}^{15}$$

is a seven-dimensional OADP variety [8, Lemma 3.2].

For the maximal type I_3 , the last three standard cocharacters span the saturated invariant lattice

$$N_3 = \mathbf{Z}e_3 \oplus \mathbf{Z}e_4 \oplus \mathbf{Z}e_5.$$

The two generators act in this basis by

$$\begin{pmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix}.$$

The four surviving weight blocks are

weight	Cox coordinates
$(0, 1, 1)$	L_{13}, L_{23}, L_{35}
$(1, 0, 1)$	L_{14}, L_{24}, L_{45}
$(1, 1, 0)$	E_1, E_2, E_5
$(1, 1, 1)$	L_{12}, L_{15}, L_{25}

and every other coordinate lies in the invariant boundary space

$$B = \mathbf{P}\langle E_3, E_4, L_{34}, Q \rangle.$$

The differences from the first weight have determinant one.

Remark 3.1 (Why the apparent double cover disappears). For type I_1 , the three visible sign cocharacters in the last three coordinates are

$$(0, 1, 1), \quad (1, 0, 1), \quad (1, 1, 0).$$

They generate the even-sum sublattice of $\mathbf{Z}^3$, of index two. Its saturation is all of $\mathbf{Z}^3$, obtained by adjoining

$$(1, 1, 1) = \frac{1}{2}((0, 1, 1) + (1, 0, 1) + (1, 1, 0)).$$

The corresponding map from the three parameter tori to the image subtorus has kernel μ_2 . The degree two seen in the unsaturated orbit coordinates counts these two parameter values; it is not a double cover of the geometric quotient. Saturation removes the kernel and makes the four weights above a unimodular tetrahedron.

Proposition 3.2 (Uniform tangent slice). *For every smooth geometric quartic del Pezzo Cox model, the rank-three subtorus above admits a tangent codimension-three slice satisfying the hypotheses of Theorem 2.1. The good slice descends to every twist of type I_0, I_1, I_2 , or I_3 over an infinite field.*

Proof. Use the standard blowup parameters

$$p_1 = (1 : 0 : 0), p_2 = (0 : 1 : 0), p_3 = (0 : 0 : 1), p_4 = (1 : 1 : 1), p_5 = (1 : a : b),$$

where $\Delta = ab(a-1)(b-1)(a-b) \neq 0$. Four exact pairs consisting of a tangent point and an orbit test point give evaluation determinants D_i and tangent smoothness minors M_i . On $\Delta \neq 0$, the first three products can vanish simultaneously only on

$$3a - b - 2 = 0$$

and either

$$Q_0(b) = 31223016b^2 - 435944529b + 1306078948 = 0$$

or $(3b - 26)(3b - 13) = 0$. On this line the primitive numerator of the fourth product is

$$-(b-1)^4(4b-17)(16b-85)(83246b^2 - 872181b + 2185995).$$

It is coprime to the two remaining equations: the four cross-evaluations are

$$\frac{69121705}{4}, \quad -\frac{4117757269}{32}, \quad \frac{7918133}{9}, \quad -\frac{272530}{9},$$

and the resultant of the two quadratics is $-48353381094689774718359510$. Thus a good geometric incidence exists on every smooth fibre. The exact witness data and the eight-branch reduction are recorded in the verification bundle.

Goodness is open in the tangent centre, the three-frame of equations, and the orbit test point. Its projection to the tangent-centre factor contains a nonempty open. Rational points on a universal torsor with one rational point are dense [8, Lemma 2.1]; hence a good centre can be chosen over k . For that centre the good frames form a nonempty open in affine frame space, which has a k -point because k is infinite. This proves descent without requiring an individual split-coordinate witness to be Galois fixed.

The type I_3 boundary and lattice are Galois stable. Up to conjugacy in the Weyl group, types I_0, I_1, I_2 are subgroups of I_3 [8, Lemma 4.2]. Restriction preserves the saturated subtorus, boundary space, unimodularity, and good incidence open. Weyl conjugation only changes the marking of the Cox coordinates. $\square$

Among the four Galois images allowed by the hereditary H^1 -condition, Proposition 3.2 is a complete classification: all four admit the same saturated unimodular window after conjugacy. We do not claim a classification for arbitrary subgroups of $W(D_5)$.

4 Proof of the level-two theorem

Proof of Theorem 1.1. Assume first that S is k -minimal. Tschinkel–Zhang’s classification places its Galois image in one of the four types above [8, Proposition 4.1 and Lemma 4.2]. Their rationality theorem for universal torsors supplies a rational projective Cox closure with dense k -points [8, Theorem 3.4]. Proposition 3.2 and Theorem 2.1 give

$$Z/T_3 \text{ rational over } k.$$

The required product identity is equivariant. If T is the Neron–Severi torus, the stable-permutation hypothesis gives $H^1(F, T) = 0$ for every extension F/k [8, Remark 2.2]. A section over $k(S)$ gives

$$S \times T \dashrightarrow \mathcal{T}, \quad (s, t) \mapsto \sigma(s)t,$$

equivariantly for T . After quotienting the anticanonical scalar, $Z \sim_k S \times T_0$ equivariantly. Therefore

$$Z/T_3 \sim_k S \times (T_0/T_3).$$

The residual torus has rank two and is rational by Voskresenskii’s theorem [9]. Hence $S \times \mathbf{A}^2$ is rational.

If S is not k -minimal, contract a k -defined union of disjoint exceptional curves and iterate. The resulting del Pezzo surface has degree at least five and a rational point, hence is k -rational by the classical rationality theorem for del Pezzo surfaces of degree at least five [5]. Thus S is already rational.

Finally, stable rationality over an infinite field supplies a rational point, and Manin’s theorem makes $\text{Pic}(S_{\bar{k}})$ stably permutation [5]. This proves the last assertion. $\square$

Proof of Corollary 1.2. Projection from the distinguished line makes each Tschinkel–Zhang cubic birational to a quartic-del-Pezzo fibration over $\mathbf{P}^{n-2}$. The generic fibre satisfies Theorem 1.1. Over the function field of the base it becomes rational after adjoining two variables. Hence $\mathbf{Q}(X)(u, v)$ is purely transcendental over $\mathbf{Q}$, which is equivalent to rationality of $X \times \mathbf{P}^2$. $\square$

Proof of Corollary 1.3. The one-stabilization theorem says that every smooth complex cubic threefold X has $X \times \mathbf{P}^1$ irrational [7, Theorem 1.1]. Corollary 1.2 gives rationality at level two, so $s(X) = 2$. If $Y = X \times \mathbf{P}^1$ were rational over $\mathbf{Q}$, it would be rational after base change to $\mathbf{C}$, a contradiction. Finally

$$\mathbf{Q}(Y \times \mathbf{A}^1) = \mathbf{Q}(X)(u, t)$$

is purely transcendental because $X \times \mathbf{P}^2$ is rational. $\square$

5 Verification boundary

The lattice saturation, generator actions, weight blocks, determinant-one window, and tangent-incidence cover use exact integer and rational arithmetic. The tracked verification bundle checks the four witnesses and the complete eight-branch cover over $\Delta \neq 0$. A separate implementation checks the lattice saturation, Galois actions, window, residual quotient action, and the displayed numerical slice minors without importing the primary certificate. The conceptual quotient theorem, equivariant generic splitting, incidence descent, and function-field passage are proved in the text rather than by the programs. The repository README records commands, hashes, and the trust boundary.

The result also recovers the stable-rationality conclusion for the four minimal Picard types with a quantitative bound. It does not reprove the rationality of universal torsors or the classification of the four types; those are the inputs from Tschinkel and Zhang. Nor does it classify every subgroup of $W(D_5)$ admitting some other unimodular Cox window.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

References

- [1] C. Ciliberto, M. Mella, and F. Russo, *Varieties with one apparent double point*, J. Algebraic Geom. 13 (2004), 475–512.
- [2] J.-L. Colliot-Thélène and J.-J. Sansuc, *La (R) -équivalence sur les tores*, Ann. Sci. École Norm. Sup. (4) 10 (1977), 175–229.
- [3] J.-L. Colliot-Thélène and J.-J. Sansuc, *La descente sur les variétés rationnelles. II*, Duke Math. J. 54 (1987), 375–492.
- [4] B. Kunyavskii, A. Skorobogatov, and M. Tsfasman, *Del Pezzo surfaces of degree four*, Mém. Soc. Math. France (N.S.) 37 (1989), 113 pp.
- [5] Yu. I. Manin, *Rational surfaces over perfect fields*, Publ. Math. IHÉS 30 (1966), 55–113.
- [6] Yu. I. Manin, *Cubic Forms*, second ed., North-Holland Mathematical Library 4, North-Holland, Amsterdam, 1986.
- [7] T. Rudd, *Irrationality of cubic threefolds after one stabilization*, Zenodo, 2026, doi:10.5281/zenodo.21909943.
- [8] Yu. Tschinkel and Zh. Zhang, *Universal torsors over quartic del Pezzo surfaces and stable rationality*, arXiv:2608.20029v1, 2026.
- [9] V. E. Voskresenskii, *On two-dimensional algebraic tori. II*, Math. USSR-Izv. 1 (1967), 691–696.